

8 ETHICS STATEMENT

There is a long history of dual-use technology in cybersecurity and there have been extensive discussion of how to weigh the benefits and risks of releasing such technologies (Rad, 2015; Silic, 2013). For instance, Silic (2013) surveys practitioners and finds that empirically practitioners agree that dual-use technology has both benefits and harms, as malicious attackers can use them for harm but good actors can use them for defense. Rad (2015) argues that while such technology can be used for harm, restrictions can hinder the benefits of the technology more than the harms, as malicious actors may simply obtain equivalent technology through alternative means such as black markets that are not available to law-abiding actors.

Here we acknowledge that the agent and the benchmark are dual-use. In this space, there have been works (Happe & Cito, 2023; Shao et al., 2024b;a; Yang et al., 2023b) that have chosen to release their code and others (Fang et al., 2024b;a;c) that have chosen to withhold the details of their research. After carefully weighing the benefits and harms of each choice, we have chosen to release our code and data and will explain our reasoning below.

In considering the harms, the concern of releasing the agent is that it may be leveraged by malicious actors to identify vulnerabilities and execute exploits on real systems (Fang et al., 2024b;a;c; Deng et al., 2023; Happe & Cito, 2023; Huang & Zhu, 2024). Current agents are not able to complete difficult cybersecurity tasks which limits the risk they pose. However, the growing capabilities of LM agents suggests that LM agents may soon substantially outclass non-LM based tools, and thereby unleash harm at a greater magnitude than existing technologies. Here, releasing the framework may accelerate development of stronger cybersecurity agents and expedite this future.

In considering the benefits, the agent can be viewed as an automated penetration testing tool. Automated penetration testing tools such as Metasploit (2024) and OWASP Nettacker (OWASP, 2024) are open-source and widely adopted with the awareness that they can be leveraged by malicious actors for attacks because the benefits vastly outweigh the risks (Abu-Dabseh & Alshammari, 2018). Here, the agent can be likened to an automated penetration testing tool as it identifies vulnerabilities and exploits them. Similarly, the benchmark would encourage development of such tools that have a similar risk-benefit profile to other automated penetration testing tools, and hence be beneficial to release.

Additionally, because related works have already openly released their code, any marginal increase in risk would be minimal. For instance, Happe & Cito (2023) release code to leverage LMs for penetration testing, arguing that attackers will use LMs and that defenders would need to prepare to defend with LMs too. Similarly Shao et al. (2024b) release code for an agent and a benchmark for CTF tasks after discussing the dual nature of AI as both a tool and a potential threat in cybersecurity. While this work has made distinct contributions, the risk profile of releasing this work is similar, and possibly less than those other works, given that alternative agents and benchmarks already exist.

Furthermore, as there has been significant interest and consideration by governments to regulate AI, we critically need more evidence and data for informed decisions and responsible regulation (Kapoor et al., 2024; Guha et al., 2023; NTIA, 2024). There have been many efforts to assess cybersecurity risk, both by government organizations such as the UK AISI (2024) and by model providers. By making our work available in a transparent fashion, we can help policymakers better understand current capabilities and risks of cybersecurity agents, when government often lacks such systematic information (NTIA, 2024). This evidence should ideally inform responsible regulatory efforts.

Finally, as scientific researchers, we believe that reproducibility and transparency are central to the AI ecosystem (of Sciences et al., 2019; Resnik & Shamoo, 2017). The reproducibility crisis affecting the sciences has affected machine learning as well, owing to mistakes and/or even fraud and fabrication (of Sciences et al., 2019; Resnik & Shamoo, 2017). While transparency in code, data, and methods is not sufficient to guarantee reproducibility (as mistakes can, of course, occur in the research process), obscurity can ensure irreproducibility. Additionally, releasing our code allows the community to build on our work, helping accelerate scientific progress.

After weighing the various factors, we choose to release our code and data publicly.

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AUTHOR CONTRIBUTIONS

Cybench was only possible because of the numerous contributions from all those involved in the effort.

Andy Zhang: Conceived of and designed the project with faculty advice, direction, and mentorship. Created initial version of codebase. Co-designed concept of subtasks. Led execution of project, including project framework, setting up organization structure, task assignments and integration process, setting up continuous integration and environment, agent creation, agent scaffolds, and subtasks. Designed the task integration process and added the first tasks with metadata and subtasks as a model for others. Led experimentation and analysis. Led writing process and wrote most of the paper.

Neil Perry: Co-designed concept of subtasks. Led design and execution of subtasks. Wrote significant portions on task categories, concepts, and analysis, and subtasks in the initial draft.

Riya Dulepet: Led design of multiple figures. Contributed significantly to creating tables and plots, running experiments, and analyzing run logs. Contributed to agent implementation. Contributed significantly to 4 tasks. Contributed significantly to writing and approving subtasks. Contributed significantly to creating and experimenting on agent scaffolds.

Joey Ji: Contributed significantly to agent development and setting up continuous integration. Contributed to creating tables and plots, running experiments, and analyzing run logs. Contributed significantly to 4 tasks. Contributed significantly to writing and approving subtasks. Led log visualization. Contributed significantly to creating and experimenting on agent scaffolds.

Celeste Menders: Contributed significantly to data analysis, visualizing logs, and analyzing tasks. Led website development effort. Contributed significantly to 5 tasks, including the most challenging in the benchmark. Contributed significantly to creating and experimenting on agent scaffolds.

Justin Lin: Contributed significantly to setting up continuous integration and environment, and agent development. Contributed to running experiments, visualizing data, and creating tables. Contributed significantly to writing and approving subtasks.

Eliot Jones: Contributed significantly to setting up continuous integration and environment, running experiments, analyzing tasks and run logs, visualizing data and creating tables. Contributed significantly to 8 tasks. Contributed to writing and approving subtasks.

Gashon Hussein: Contributed significantly to setting up continuous integration and environment, and agent development.

Samantha Liu: Led effort to parse and interpret first blood data and wrote the first draft of that appendix. Contributed to running experiments, agent implementation, analyzing run logs, and table creation. Contributed significantly to 4 tasks.

Donovan Jasper: Contributed significantly to writing and approving subtasks, and analyzing tasks and competitions. Contributed significantly to 4 tasks.

Pura Peetathawatchai: Contributed significantly to writing and approving subtasks. Contributed significantly to 2 tasks.

Ari Glenn: Contributed significantly to 4 tasks. Hosted sessions to help others with tasks.

Vikram Sivashankar: Contributed significantly to 4 tasks. Contributed to agent development.

Daniel Zamoshchin: Contributed significantly to 4 tasks that were especially difficult.

Leo Glikbarg: Contributed significantly to 3 tasks. Contributed to writing and approving subtasks.

Derek Askaryar: Contributed significantly to 3 tasks.

Mike Yang: Contributed significantly to 3 tasks.

Teddy Zhang: Contributed significantly to 2 tasks.

Rishi Alluri: Contributed significantly to 2 tasks.

Nathan Tran: Contributed significantly to 2 tasks.

Rinnara Sangpisit: Contributed significantly to 2 tasks.

Polycarpus Yiorkadjis: Contributed significantly to 2 tasks.

Kenny Osele: Contributed significantly to 1 task.

Gautham Raghuathi: Contributed significantly to 1 task.

Dan Boneh: Provided overall guidance on the project, especially in cybersecurity, including project conception, direction, and framing. Provided overall feedback and guidance on the paper.

Daniel E. Ho: Led initial discussions for project formation and ideation. Provided overall guidance on the project, especially in policy, including project conception, direction, and framing. Provided overall feedback and guidance on the paper.

Percy Liang: Led initial discussions for project formation and ideation. Led and managed the overall project. Led project conception, scoping, and direction. Provided overall guidance on the project including project conception, direction, and framing. Provided guidance on the agent and benchmark design, organizational structure, and code structure. Provided overall feedback and guidance on the paper.

REFERENCES

- Farah Abu-Dabseh and Esraa Alshammari. Automated penetration testing: An overview. In *The 4th international conference on natural language computing*, Copenhagen, Denmark, pp. 121–129, 2018.
- Anthropic. Claude 3.5 sonnet, 2024a. URL <https://www.anthropic.com/news/claude-3-5-sonnet>.
- Anthropic. Claude 3, 2024b. URL <https://www-cdn.anthropic.com/f2986af8d052f26236f6251da62d16172cfabd6e/claude-3-model-card.pdf>.
- Anthropic. Models - anthropic. <https://docs.anthropic.com/en/docs/about-claude/models#model-comparison>, 2024c. Accessed: 2024-08-13.
- Anthropic. Claude 3 models, 2024d. URL <https://docs.anthropic.com/en/docs/about-claude/models>. Accessed: 2024-08-10.
- Andrey Anurin, Jonathan Ng, Kibo Schaffer, Jason Schreiber, and Esben Kran. Catastrophic cyber capabilities benchmark (3cb): Robustly evaluating LLM agent cyber offense capabilities, 2024. URL <https://arxiv.org/abs/2410.09114>.
- Manish Bhatt, Sahana Chennabasappa, Yue Li, Cyrus Nikolaidis, Daniel Song, Shengye Wan, Faizan Ahmad, Cornelius Aschermann, Yaohui Chen, Dhaval Kapil, David Molnar, Spencer Whitman, and Joshua Saxe. Cyberseceval 2: A wide-ranging cybersecurity evaluation suite for large language models, 2024. URL <https://arxiv.org/abs/2404.13161>.
- ctfTime. Ctf competition participants, 2023. URL <https://ctftime.org/ctfs>. Accessed: 2024-06-26.
- ctfTime Glacier. Glacier ctf 2023 competition, 2023. URL <https://ctftime.org/event/1992/>. Accessed: 2024-06-25.
- Gelei Deng, Yi Liu, Víctor Mayoral-Vilches, Peng Liu, Yuekang Li, Yuan Xu, Tianwei Zhang, Yang Liu, Martin Pinzger, and Stefan Rass. Pentestgpt: An LLM-empowered automatic penetration testing tool, 2023.
- Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, Anirudh Goyal, Anthony Hartshorn, Aobo Yang, Archi Mitra, Archie Sravankumar, Artem Korenev, Arthur Hinsvark, Arun Rao, Aston Zhang, Aurelien Rodriguez, Austen Gregerson, Ava Spataru, Baptiste Roziere, Bethany Biron, Binh Tang, Bobbie Chern, Charlotte Caucheteux, Chaya Nayak, Chloe Bi, Chris Marra, Chris McConnell, Christian Keller, Christophe Touret, Chunyang Wu, Corinne Wong, Cristian Canton Ferrer, Cyrus Nikolaidis, Damien Allonsius, Daniel Song, Danielle Pintz, Danny Livshits, David Esiobu, Dhruv Choudhary, Dhruv Mahajan, Diego Garcia-Olano, Diego Perino, Dieuwke Hupkes, Egor Lakomkin, Ehab AlBadawy, Elina Lobanova, Emily Dinan, Eric Michael Smith, Filip Radenovic, Frank Zhang, Gabriel Synnaeve, Gabrielle Lee, Georgia Lewis Anderson, Graeme Nail, Gregoire Mialon, Guan Pang, Guillem Cucurell, Hailey Nguyen, Hannah

Korevaar, Hu Xu, Hugo Touvron, Iliyan Zarov, Imanol Arrieta Ibarra, Isabel Kloumann, Ishan Misra, Ivan Evtimov, Jade Copet, Jaewon Lee, Jan Geffert, Jana Vranes, Jason Park, Jay Mahadeokar, Jeet Shah, Jelmer van der Linde, Jennifer Billock, Jenny Hong, Jenya Lee, Jeremy Fu, Jianfeng Chi, Jianyu Huang, Jiawen Liu, Jie Wang, Jiecao Yu, Joanna Bitton, Joe Spisak, Jongsoo Park, Joseph Rocca, Joshua Johnstun, Joshua Saxe, Junteng Jia, Kalyan Vasuden Alwala, Kartikeya Upasani, Kate Plawiak, Ke Li, Kenneth Heafield, Kevin Stone, Khalid El-Arini, Krithika Iyer, Kshitiz Malik, Kuenley Chiu, Kunal Bhalla, Lauren Rantala-Yearly, Laurens van der Maaten, Lawrence Chen, Liang Tan, Liz Jenkins, Louis Martin, Lovish Madaan, Lubo Malo, Lukas Blecher, Lukas Landzaat, Luke de Oliveira, Madeline Muzzi, Mahesh Pasupuleti, Manat Singh, Manohar Paluri, Marcin Kardas, Mathew Oldham, Mathieu Rita, Maya Pavlova, Melanie Kambadur, Mike Lewis, Min Si, Mitesh Kumar Singh, Mona Hassan, Naman Goyal, Narjes Torabi, Nikolay Bashlykov, Nikolay Bogoychev, Niladri Chatterji, Olivier Duchenne, Onur Çelebi, Patrick Alrassy, Pengchuan Zhang, Pengwei Li, Petar Vasic, Peter Weng, Prajjwal Bhargava, Pratik Dubal, Praveen Krishnan, Punit Singh Koura, Puxin Xu, Qing He, Qingxiao Dong, Ragavan Srinivasan, Raj Ganapathy, Ramon Calderer, Ricardo Silveira Cabral, Robert Stojnic, Roberta Raileanu, Rohit Girdhar, Rohit Patel, Romain Sauvestre, Ronnie Polidoro, Roshan Sumbaly, Ross Taylor, Ruan Silva, Rui Hou, Rui Wang, Saghar Hosseini, Sahana Chennabasappa, Sanjay Singh, Sean Bell, Seohyun Sonia Kim, Sergey Edunov, Shaoliang Nie, Sharan Narang, Sharath Raparthy, Sheng Shen, Shengye Wan, Shruti Bhosale, Shun Zhang, Simon Vandenhende, Soumya Batra, Spencer Whitman, Sten Sootla, Stephane Collot, Suchin Gururangan, Sydney Borodinsky, Tamar Herman, Tara Fowler, Tarek Sheasha, Thomas Georgiou, Thomas Scialom, Tobias Speckbacher, Todor Mihaylov, Tong Xiao, Ujjwal Karn, Vedanuj Goswami, Vibhor Gupta, Vignesh Ramanathan, Viktor Kerkez, Vincent Gonguet, Virginie Do, Vish Vogeti, Vladan Petrovic, Weiwei Chu, Wenhan Xiong, Wenyin Fu, Whitney Meers, Xavier Martinet, Xiaodong Wang, Xiaoqing Ellen Tan, Xinfeng Xie, Xuchao Jia, Xuwei Wang, Yaelle Goldschlag, Yashesh Gaur, Yasmine Babaei, Yi Wen, Yiwen Song, Yuchen Zhang, Yue Li, Yuning Mao, Zacharie Delpierre Coudert, Zheng Yan, Zhengxing Chen, Zoe Papakipos, Aaditya Singh, Aaron Grattafiori, Abha Jain, Adam Kelsey, Adam Shajnfeld, Adithya Gangidi, Adolfo Victoria, Ahuva Goldstand, Ajay Menon, Ajay Sharma, Alex Boesenberg, Alex Vaughan, Alexei Baevski, Allie Feinstein, Amanda Kallet, Amit Sangani, Anam Yunus, Andrei Lupu, Andres Alvarado, Andrew Caples, Andrew Gu, Andrew Ho, Andrew Poulton, Andrew Ryan, Ankit Ramchandani, Annie Franco, Aparajita Saraf, Arkabandhu Chowdhury, Ashley Gabriel, Ashwin Bharambe, Assaf Eisenman, Azadeh Yazdan, Beau James, Ben Maurer, Benjamin Leonhardi, Bernie Huang, Beth Loyd, Beto De Paola, Bhargavi Paranjape, Bing Liu, Bo Wu, Boyu Ni, Braden Hancock, Bram Wasti, Brandon Spence, Brani Stojkovic, Brian Gamido, Britt Montalvo, Carl Parker, Carly Burton, Catalina Mejia, Changan Wang, Changkyu Kim, Chao Zhou, Chester Hu, Ching-Hsiang Chu, Chris Cai, Chris Tindal, Christoph Feichtenhofer, Damon Civin, Dana Beaty, Daniel Kreymer, Daniel Li, Danny Wyatt, David Adkins, David Xu, Davide Testuggine, Delia David, Devi Parikh, Diana Liskovich, Didem Foss, Dingkan Wang, Duc Le, Dustin Holland, Edward Dowling, Eissa Jamil, Elaine Montgomery, Eleonora Presani, Emily Hahn, Emily Wood, Erik Brinkman, Esteban Arcaute, Evan Dunbar, Evan Smothers, Fei Sun, Felix Kreuk, Feng Tian, Firat Ozgenel, Francesco Caggioni, Francisco Guzmán, Frank Kanayet, Frank Seide, Gabriela Medina Florez, Gabriella Schwarz, Gada Badeer, Georgia Swee, Gil Halpern, Govind Thattai, Grant Herman, Grigory Sizov, Guangyi, Zhang, Guna Lakshminarayanan, Hamid Shojanazeri, Han Zou, Hannah Wang, Hanwen Zha, Haroun Habeeb, Harrison Rudolph, Helen Suk, Henry Aspegren, Hunter Goldman, Igor Molybog, Igor Tufanov, Irina-Elena Veliche, Itai Gat, Jake Weissman, James Geboski, James Kohli, Japhet Asher, Jean-Baptiste Gaya, Jeff Marcus, Jeff Tang, Jennifer Chan, Jenny Zhen, Jeremy Reizenstein, Jeremy Teboul, Jessica Zhong, Jian Jin, Jingyi Yang, Joe Cummings, Jon Carvill, Jon Shepard, Jonathan McPhie, Jonathan Torres, Josh Ginsburg, Junjie Wang, Kai Wu, Kam Hou U, Karan Saxena, Karthik Prasad, Kartikay Khandelwal, Katayoun Zand, Kathy Matosich, Kaushik Veeraraghavan, Kelly Michelena, Keqian Li, Kun Huang, Kunal Chawla, Kushal Lakhotia, Kyle Huang, Lailin Chen, Lakshya Garg, Lavender A, Leandro Silva, Lee Bell, Lei Zhang, Liangpeng Guo, Licheng Yu, Liron Moshkovich, Luca Wehrstedt, Madian Khabza, Manav Avalani, Manish Bhatt, Maria Tsimpoukelli, Martynas Mankus, Matan Hasson, Matthew Lennie, Matthias Reso, Maxim Groshev, Maxim Naumov, Maya Lathi, Meghan Keenally, Michael L. Seltzer, Michal Valko, Michelle Restrepo, Mihir Patel, Mik Vyatskov, Mikayel Samvelyan, Mike Clark, Mike Macey, Mike Wang, Miquel Jubert Hermoso, Mo Metanat, Mohammad Rastegari, Munish Bansal, Nandhini Santhanam, Natascha Parks, Natasha White, Navyata Bawa, Nayan Singhal, Nick Egebo, Nicolas Usunier, Nikolay Pavlovich Laptev, Ning Dong,

Ning Zhang, Norman Cheng, Oleg Chernoguz, Olivia Hart, Omkar Salpekar, Ozlem Kalinli, Parkin Kent, Parth Parekh, Paul Saab, Pavan Balaji, Pedro Rittner, Philip Bontrager, Pierre Roux, Piotr Dollar, Polina Zvyagina, Prashant Ratanchandani, Pritish Yuvraj, Qian Liang, Rachad Alao, Rachel Rodriguez, Rafi Ayub, Raghotham Murthy, Raghu Nayani, Rahul Mitra, Raymond Li, Rebekkah Hogan, Robin Battey, Rocky Wang, Rohan Maheswari, Russ Howes, Ruty Rinott, Sai Jayesh Bondu, Samyak Datta, Sara Chugh, Sara Hunt, Sargun Dhillon, Sasha Sidorov, Satadru Pan, Saurabh Verma, Seiji Yamamoto, Sharadh Ramaswamy, Shaun Lindsay, Shaun Lindsay, Sheng Feng, Shenghao Lin, Shengxin Cindy Zha, Shiva Shankar, Shuqiang Zhang, Shuqiang Zhang, Sinong Wang, Sneha Agarwal, Soji Sajuyigbe, Soumith Chintala, Stephanie Max, Stephen Chen, Steve Kehoe, Steve Satterfield, Sudarshan Govindaprasad, Sumit Gupta, Sungmin Cho, Sunny Virk, Suraj Subramanian, Sy Choudhury, Sydney Goldman, Tal Remez, Tamar Glaser, Tamara Best, Thilo Kohler, Thomas Robinson, Tianhe Li, Tianjun Zhang, Tim Matthews, Timothy Chou, Tzook Shaked, Varun Vontimitta, Victoria Ajayi, Victoria Montanez, Vijai Mohan, Vinay Satish Kumar, Vishal Mangla, Vlad Ionescu, Vlad Poenaru, Vlad Tiberiu Mihailescu, Vladimir Ivanov, Wei Li, Wenchen Wang, Wenwen Jiang, Wes Bouaziz, Will Constable, Xiaocheng Tang, Xiaofang Wang, Xiaojuan Wu, Xiaolan Wang, Xide Xia, Xilun Wu, Xinbo Gao, Yanjun Chen, Ye Hu, Ye Jia, Ye Qi, Yenda Li, Yilin Zhang, Ying Zhang, Yossi Adi, Youngjin Nam, Yu, Wang, Yuchen Hao, Yundi Qian, Yuzi He, Zach Rait, Zachary DeVito, Zef Rosnbrick, Zhaoduo Wen, Zhenyu Yang, and Zhiwei Zhao. The llama 3 herd of models, 2024. URL <https://arxiv.org/abs/2407.21783>.

Aparna Elangovan, Jiayuan He, and Karin Verspoor. Memorization vs. generalization: quantifying data leakage in nlp performance evaluation. *arXiv preprint arXiv:2102.01818*, 2021.

Richard Fang, Rohan Bindu, Akul Gupta, and Daniel Kang. LLM agents can autonomously exploit one-day vulnerabilities. *arXiv preprint arXiv:2404.08144*, 2024a.

Richard Fang, Rohan Bindu, Akul Gupta, Qiusi Zhan, and Daniel Kang. LLM agents can autonomously hack websites, 2024b. URL <https://arxiv.org/abs/2402.06664>.

Richard Fang, Rohan Bindu, Akul Gupta, Qiusi Zhan, and Daniel Kang. Teams of LLM agents can exploit zero-day vulnerabilities, 2024c. URL <https://arxiv.org/abs/2406.01637>.

Google. Gemini 1.5 pro. <https://ai.google.dev/gemini-api/docs/models/gemini#gemini-1.5-pro>, 2024a. Accessed: 2024-08-13.

Google. Gemini 1.5, 2024b. URL <https://arxiv.org/pdf/2403.05530>.

Neel Guha, Christie Lawrence, Lindsey A Gailmard, Kit Rodolfa, Faiz Surani, Rishi Bommasani, Inioluwa Raji, Mariano-Florentino Cuéllar, Colleen Honigsberg, Percy Liang, et al. Ai regulation has its own alignment problem: The technical and institutional feasibility of disclosure, registration, licensing, and auditing. *George Washington Law Review*, Forthcoming, 2023.

Hack The Box. Cyber apocalypse 2024, 2024. URL <https://github.com/hackthebox/cyber-apocalypse-2024>. Accessed: 2024-05-20.

Andreas Happe and Jürgen Cito. Getting pwn’d by ai: Penetration testing with large language models. In *Proceedings of the 31st ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering, ESEC/FSE ’23*. ACM, November 2023. doi: 10.1145/3611643.3613083. URL <http://dx.doi.org/10.1145/3611643.3613083>.

HKCERT. Hkcert 2023 ctf competition participants, 2023. URL <https://www.hkcert.org/press-centre/hong-kong-cyber-security-new-generation-capture-the-flag-challenge-2023#:~:text=As%20one%20of%20the%20world's,forming%20more%20than%20500%20teams>. Accessed: 2024-06-26.

HKCert CTF. Ctf challenges, 2023. URL <https://github.com/hkcert-ctf/CTF-Challenges>. Accessed: 2024-05-20.

hkcertCTF. Hkcert ctf competition, 2023. URL <https://ctf.hkcert.org/index.html>. Accessed: 2024-06-25.